

3. Continuous Delivery with Google Cloud Deploy

Dashboard – qwiklabs-gcp-04-93e61e9975d5 x +

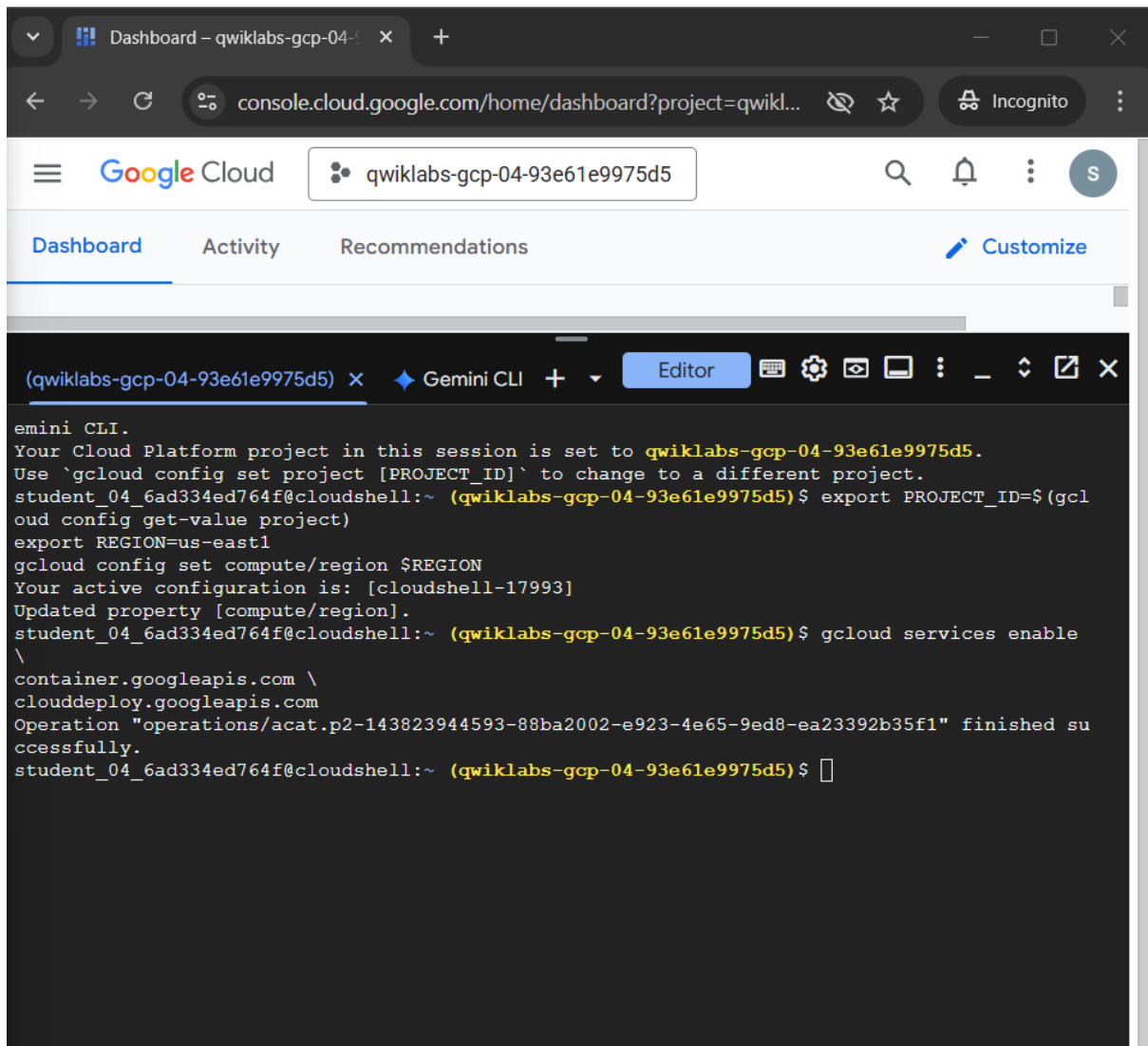
← → ↻ console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-93e61e9975d5 Incognito

Google Cloud qwiklabs-gcp-04-93e61e9975d5

Dashboard Activity Recommendations Customize

(qwiklabs-gcp-04-93e61e9975d5) x Gemini CLI + Editor

```
emini CLI.
Your Cloud Platform project in this session is set to qwiklabs-gcp-04-93e61e9975d5.
Use `gcloud config set project [PROJECT_ID]` to change to a different project.
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5) $ export PROJECT_ID=$(gcloud config get-value project)
export REGION=us-east1
gcloud config set compute/region $REGION
Your active configuration is: [cloudshell-17993]
Updated property [compute/region].
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5) $
```



The screenshot shows a web browser window displaying the Google Cloud console. The address bar shows the URL `console.cloud.google.com/home/dashboard?project=qwikl...`. The page header includes the Google Cloud logo, the project ID `qwiklabs-gcp-04-93e61e9975d5`, and navigation links for `Dashboard`, `Activity`, and `Recommendations`. A `Customize` link is also present. Below the header, a terminal window is open, showing the following text:

```
(qwiklabs-gcp-04-93e61e9975d5) x  Gemini CLI  +  Editor  [Icons]
emini CLI.
Your Cloud Platform project in this session is set to qwiklabs-gcp-04-93e61e9975d5.
Use `gcloud config set project [PROJECT_ID]` to change to a different project.
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)$ export PROJECT_ID=$(gcl
oud config get-value project)
export REGION=us-east1
gcloud config set compute/region $REGION
Your active configuration is: [cloudshell-17993]
Updated property [compute/region].
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)$ gcloud services enable
\
container.googleapis.com \
clouddeploy.googleapis.com
Operation "operations/acat.p2-143823944593-88ba2002-e923-4e65-9ed8-ea23392b35f1" finished su
ccessfully.
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)$
```

Dashboard – qwiklabs-gcp-04-93e61e9975d5

console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-93e61e9975d5

Incognito

Google Cloud

qwiklabs-gcp-04-93e61e9975d5

Dashboard Activity Recommendations Customize

(qwiklabs-gcp-04-93e61e9975d5) x Gemini CLI + Editor

clouddeploy.googleapis.com
Operation "operations/acat.p2-143823944593-88ba2002-e923-4e65-9ed8-ea23392b35f1" finished successfully.
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)\$ gcloud container clusters create test --node-locations=us-east1-d --num-nodes=1 --async
gcloud container clusters create staging --node-locations=us-east1-d --num-nodes=1 --async
gcloud container clusters create prod --node-locations=us-east1-d --num-nodes=1 --async
Note: Your Pod address range ('--cluster-ipv4-cidr') can accommodate at most 1008 node(s).
NAME: test
TYPE:
LOCATION: us-east1
TARGET:
STATUS_MESSAGE:
STATUS: PROVISIONING
START_TIME:
END_TIME:
Note: Your Pod address range ('--cluster-ipv4-cidr') can accommodate at most 1008 node(s).
NAME: staging
TYPE:
LOCATION: us-east1
TARGET:
STATUS_MESSAGE:
STATUS: PROVISIONING
START_TIME:
END_TIME:
Note: Your Pod address range ('--cluster-ipv4-cidr') can accommodate at most 1008 node(s).
NAME: prod
TYPE:
LOCATION: us-east1
TARGET:
STATUS_MESSAGE:
STATUS: PROVISIONING
START_TIME:
END_TIME:
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)\$

Dashboard – qwiklabs-gcp-04-93e61e9975d5

console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-93e61e9975d5

Google Cloud

qwiklabs-gcp-04-93e61e9975d5

Dashboard Activity Recommendations Customize

(qwiklabs-gcp-04-93e61e9975d5) Gemini CLI Editor

gcloud container clusters create prod --node-locations=us-east1-d --num-nodes=1 --async
Note: Your Pod address range (--cluster-ipv4-cidr) can accommodate at most 1008 node(s).
NAME: test
TYPE:
LOCATION: us-east1
TARGET:
STATUS_MESSAGE:
STATUS: PROVISIONING
START_TIME:
END_TIME:
Note: Your Pod address range (--cluster-ipv4-cidr) can accommodate at most 1008 node(s).
NAME: staging
TYPE:
LOCATION: us-east1
TARGET:
STATUS_MESSAGE:
STATUS: PROVISIONING
START_TIME:
END_TIME:
Note: Your Pod address range (--cluster-ipv4-cidr) can accommodate at most 1008 node(s).
NAME: prod
TYPE:
LOCATION: us-east1
TARGET:
STATUS_MESSAGE:
STATUS: PROVISIONING
START_TIME:
END_TIME:
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)\$ gcloud container clusters list --format="csv(name,status)"
name,status
prod,PROVISIONING
staging,PROVISIONING
test,PROVISIONING
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)\$

student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)\$ gcloud services enable artifactregistry.googleapis.com
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)\$ gcloud artifacts repositories create web-app \
--description="Image registry for tutorial web app" \
--repository-format=docker \
--location=\$REGION
Create request issued for: [web-app]
Waiting for operation [projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/operations/500a5e39-c817-4457-a266-669bef64b810] to complete...done.
Created repository [web-app].
student_04_6ad334ed764f@cloudshell:~ (qwiklabs-gcp-04-93e61e9975d5)\$

Dashboard - qwiklabs-gcp-04-93e61e9975d5 x +

console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-93e61e9975d5

Google Cloud qwiklabs-gcp-04-93e61e9975d5

Dashboard Activity Recommendations Customize

(qwiklabs-gcp-04-93e61e9975d5) x Gemini CLI + Editor

```
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5)$ gcloud services enable cloudbuild.googleapis.com
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5)$ gsutil mb -p $PROJECT_ID gs://$PROJECT_ID_cloudbuild
Creating gs://qwiklabs-gcp-04-93e61e9975d5_cloudbuild/...
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5)$ cd web
skaffold build --interactive=false \
--default-repo $REGION-docker.pkg.dev/$PROJECT_ID/web-app \
--file-output artifacts.json
cd ..
Generating tags...
- leeroy-web -> us-east1-docker.pkg.dev/qwiklabs-gcp-04-93e61e9975d5/web-app/leeroy-web:c3cae80
- leeroy-app -> us-east1-docker.pkg.dev/qwiklabs-gcp-04-93e61e9975d5/web-app/leeroy-app:c3cae80
Checking cache...
- leeroy-web: Not found. Building
- leeroy-app: Not found. Building
Starting build...
Building 2 artifacts in parallel
Building [leeroy-web]...
Pushing code to gs://qwiklabs-gcp-04-93e61e9975d5_cloudbuild/source/qwiklabs-gcp-04-93e61e9975d5-2da244b0-41b5-48b2-bb68-14115efecf20.tar.gz
Logs are available at
https://storage.cloud.google.com/qwiklabs-gcp-04-93e61e9975d5_cloudbuild/log-a08d2fe3-b6b4-4f00-8382-23f07a215c2d.txt
starting build "a08d2fe3-b6b4-4f00-8382-23f07a215c2d"

FETCHSOURCE
Fetching storage object: gs://qwiklabs-gcp-04-93e61e9975d5_cloudbuild/source/qwiklabs-gcp-04-93e61e9975d5-2da244b0-41b5-48b2-bb68-14115efecf20.tar.gz#1760363028234840
Copying gs://qwiklabs-gcp-04-93e61e9975d5_cloudbuild/source/qwiklabs-gcp-04-93e61e9975d5-2da244b0-41b5-48b2-bb68-14115efecf20.tar.gz#1760363028234840...
/ [1 files][ 981.0 B/ 981.0 B]
```

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Dashboard - qwiklabs-gcp-04-93e61e9975d5

console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-93e61e9975d5

Google Cloud qwiklabs-gcp-04-93e61e9975d5

Dashboard Activity Recommendations Customize

(qwiklabs-gcp-04-93e61e9975d5) Gemini CLI Editor

```
Step 1/7 : FROM golang:1.12.9-alpine3.10 as builder
1.12.9-alpine3.10: Pulling from library/golang
9d48c3bd43c5: Pulling fs layer
7f94eaf8af20: Pulling fs layer
9fe9984849c1: Pulling fs layer
cf0db633a67d: Pulling fs layer
0f7136d71739: Pulling fs layer
cf0db633a67d: Waiting
0f7136d71739: Waiting
9fe9984849c1: Verifying Checksum
9fe9984849c1: Download complete
9d48c3bd43c5: Verifying Checksum
9d48c3bd43c5: Download complete
7f94eaf8af20: Verifying Checksum
7f94eaf8af20: Download complete
9d48c3bd43c5: Pull complete
0f7136d71739: Verifying Checksum
0f7136d71739: Download complete
7f94eaf8af20: Pull complete
9fe9984849c1: Pull complete
cf0db633a67d: Verifying Checksum
cf0db633a67d: Download complete
cf0db633a67d: Pull complete
0f7136d71739: Pull complete
Digest: sha256:e0660b4f1e68e0d408420acb874b396fc6dd25e7c1d03ad36e7d6d1155a4dff6
Status: Downloaded newer image for golang:1.12.9-alpine3.10
---> e0d646523991
Step 2/7 : COPY web.go .
---> cfe62aa9cad5
Step 3/7 : RUN go build -o /web .
---> Running in 4f9705221831
Removing intermediate container 4f9705221831
---> 2927794ee11b
Step 4/7 : FROM alpine:3.10
3.10: Pulling from library/alpine
```

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Dashboard - qwiklabs-gcp-04-93e61e9975d5

console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-93e61e9975d5

Google Cloud qwiklabs-gcp-04-93e61e9975d5

Dashboard Activity Recommendations Customize

(qwiklabs-gcp-04-93e61e9975d5) Gemini CLI Editor

```
package: us-east1-docker.pkg.dev/qwiklabs-gcp-04-93e61e9975d5/web-app/leeroy-web
tags:
- c3cae80
updateTime: '2025-10-13T13:44:12.373743Z'
version: sha256:13b9834adf669cb8b1345f0fa9470926d99e39a39de2cfc1af2aaba426d2737e
---
createTime: '2025-10-13T13:44:27.433055Z'
metadata:
  buildTime: null
  imageSizeBytes: '5411'
  mediaType: application/vnd.oci.image.manifest.v1+json
  name: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/repositories/web-app/docker
Images/leeroy-web@sha256:dd3821ea572f0f7e161bcl3a3288182c1b35f4662d9b0a94466eee16d66a777
package: us-east1-docker.pkg.dev/qwiklabs-gcp-04-93e61e9975d5/web-app/leeroy-web
tags: []
updateTime: '2025-10-13T13:44:27.433055Z'
version: sha256:dd3821ea572f0f7e161bcl3a3288182c1b35f4662d9b0a94466eee16d66a777
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5) $ cat web/artifacts.json | jq
{
  "builds": [
    {
      "imageName": "leeroy-web",
      "tag": "us-east1-docker.pkg.dev/qwiklabs-gcp-04-93e61e9975d5/web-app/leeroy-web:c3cae80@sha256:13b9834adf669cb8b1345f0fa9470926d99e39a39de2cfc1af2aaba426d2737e"
    },
    {
      "imageName": "leeroy-app",
      "tag": "us-east1-docker.pkg.dev/qwiklabs-gcp-04-93e61e9975d5/web-app/leeroy-app:c3cae80@sha256:a3fe1f6cf762db0c9e04378fe5bdb4f2270eabcf11163e1fd92b09561943afbd"
    }
  ]
}
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5) $
```

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```
93e61e9975d5) $ gcloud config set deploy/region $REGION
cp clouddeploy-config/delivery-pipeline.yaml.template clouddeploy-config/delivery-pipeline.yaml
Updated property [deploy/region].
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5) $ gcloud container clusters list --format="csv(name,status)"
name,status
prod,RUNNING
staging,RUNNING
test,RUNNING
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5) $
```

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Dashboard - qwiklabs-gcp-04-93e61e9975d5

console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-93e61e9975d5&pli=1&cloudshell=true

Google Cloud qwiklabs-gcp-04-93e61e9975d5 Search (/) for resources, docs, products, and more

Dashboard Activity Recommendations Customize

CLOUD SHELL Terminal (qwiklabs-gcp-04-93e61e9975d5) Gemini CLI + Open Editor

```
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5)$ CONTEXTS=("test" "staging" "prod")
for CONTEXT in $(CONTEXTS[@]);
do
  gcloud container clusters get-credentials $(CONTEXT) --region $(REGION)
  kubectl config rename-context gke_${PROJECT_ID}_${REGION}_${CONTEXT} $(CONTEXT)
done
Fetching cluster endpoint and auth data.
kubeconfig entry generated for test.
Context "gke_qwiklabs-gcp-04-93e61e9975d5_us-east1_test" renamed to "test".
Fetching cluster endpoint and auth data.
kubeconfig entry generated for staging.
Context "gke_qwiklabs-gcp-04-93e61e9975d5_us-east1_staging" renamed to "staging".
Fetching cluster endpoint and auth data.
kubeconfig entry generated for prod.
Context "gke_qwiklabs-gcp-04-93e61e9975d5_us-east1_prod" renamed to "prod".
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5)$ for CONTEXT in $(CONTEXTS[@])
do
  kubectl --context $(CONTEXT) apply -f kubernetes-config/web-app-namespace.yaml
done
namespace/web-app created
namespace/web-app created
namespace/web-app created
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5)$ for CONTEXT in $(CONTEXTS[@])
do
  envsubst < clouddeploy-config/target-SCONTEXT.yaml.template > clouddeploy-config/target-SCONTEXT.yaml
  gcloud beta deploy apply --file clouddeploy-config/target-SCONTEXT.yaml
done
Waiting for the operation on resource projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/targets/test...done.
Created Cloud Deploy resource: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/targets/test.
Waiting for the operation on resource projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/targets/staging...done.
Created Cloud Deploy resource: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/targets/staging.
Waiting for the operation on resource projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/targets/prod...done.
Created Cloud Deploy resource: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/targets/prod.
```

student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5)\$ cat clouddeploy-config/target-test.yaml

```
# Copyright 2021 Google LLC
# Licensed under the Apache License, Version 2.0 (the "License");
# you may not use this file except in compliance with the License.
# You may obtain a copy of the License at
#
# http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the license is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.

apiVersion: deploy.cloud.google.com/v1beta1
kind: Target
metadata:
  name: test
  description: test cluster
gke:
  cluster: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/clusters/test
```

student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5)\$ cat clouddeploy-config/target-prod.yaml

```
# Copyright 2021 Google LLC
# Licensed under the Apache License, Version 2.0 (the "License");
# you may not use this file except in compliance with the License.
# You may obtain a copy of the License at
#
# http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the license is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
```

Dashboard - qwiklabs-gcp-04-93e61e9975d5

console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-93e61e9975d5&pli=1&cloudshell=true

Google Cloud qwiklabs-gcp-04-93e61e9975d5 Search (/) for resources, docs, products, and more

Dashboard Activity Recommendations Customize

CLOUD SHELL Terminal (qwiklabs-gcp-04-93e61e9975d5) Gemini CLI + Open Editor

```
- RENDER
- DEPLOY
- VERIFY
- PREDEPLOY
- POSTDEPLOY

gke:
  cluster: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/clusters/test
  name: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/targets/test
  targetId: test
  uid: 4b75220d3dc24a5d976c2fed1998d8a7
  updateTime: '2025-10-13T13:47:49.138946779z'
  createTime: '2025-10-13T13:48:00.035041605z'
  description: prod cluster
  etag: fa2c44801ec10792
  executionConfig:
    artifactStorage: gs://us-east1.deploy-artifacts.qwiklabs-gcp-04-93e61e9975d5.appspot.com
    defaultPool:
      artifactStorage: gs://us-east1.deploy-artifacts.qwiklabs-gcp-04-93e61e9975d5.appspot.com
      serviceAccount: 143823944593-compute@developer.gserviceaccount.com
      executionTimeout: 3600s
      serviceAccount: 143823944593-compute@developer.gserviceaccount.com
    usages:
      - RENDER
      - DEPLOY
      - VERIFY
      - PREDEPLOY
      - POSTDEPLOY

gke:
  cluster: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/clusters/prod
  name: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/targets/prod
  requireApproval: true
  targetId: prod
  uid: bdecc8deb6884a838e91b10e876b099a
student_04_ead334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5) $
```

24°C Clear 6:49 PM 13-Oct-2025

Dashboard - qwiklabs-gcp-04-... x +

console.cloud.google.com/home/dashboard?project=qwikl... Incognito

Google Cloud qwiklabs-gcp-04-93e61e9975d5

Dashboard Activity Recommendations Customize

(quiklabs-gcp-04-93e61e9975d5) x Gemini CLI + Editor

```
id: stable
state: IN_PROGRESS
state: IN_PROGRESS
targetId: staging
uid: 223b1c779b13422390c6f3c15b291077
---
approvalState: DOES_NOT_NEED_APPROVAL
createTime: '2025-10-13T13:52:55.316741Z'
deployEndTime: '2025-10-13T13:53:35.041573Z'
deployStartTime: '2025-10-13T13:53:08.573794551Z'
deployingBuild: projects/143823944593/locations/us-east1/builds/97d5c4ae-3c58-42f6-a808-41efb1120ba6
enqueueTime: '2025-10-13T13:53:08.222239Z'
etag: 17feab07705c0b59
name: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/deliveryPipelines/web-app/releases/web-app-001/rollouts/web-app-001-to-test-0001
phases:
- deploymentJobs:
  deploymentJob:
    deployJob: {}
    id: deploy
    jobRun: projects/143823944593/locations/us-east1/deliveryPipelines/web-app/releases/web-app-001/rollouts/web-app-001-to-test-0001/jobRuns/b967eb05-f3e1-4806-85c1-344f37b559f7
    state: SUCCEEDED
    verifyJob:
      id: verify
      state: DISABLED
      verifyJob: {}
  id: stable
  state: SUCCEEDED
state: SUCCEEDED
targetId: test
uid: f657cf4351e84b6b97396c6a6dcea6c7
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (quiklabs-gcp-04-93e61e9975d5) $
```

6:54 PM 13 Oct 2025

Dashboard - qwiklabs-gcp-04-93e61e9975d5 x +

console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-93e61e9975d5 Incognito

Google Cloud qwiklabs-gcp-04-93e61e9975d5

Dashboard Activity Recommendations Customize

(qwiklabs-gcp-04-93e61e9975d5) x Gemini CLI + Editor

```
approvalState: DOES_NOT_NEED_APPROVAL
createTime: '2025-10-13T13:52:55.316741Z'
deployEndTime: '2025-10-13T13:53:35.041573Z'
deployStartTime: '2025-10-13T13:53:08.573794551Z'
deployingBuild: projects/143823944593/locations/us-east1/builds/97d5c4ae-3c58-42f6-a808-41efb1120ba6
enqueueTime: '2025-10-13T13:53:08.222239Z'
etag: 17feab07705c0b59
name: projects/qwiklabs-gcp-04-93e61e9975d5/locations/us-east1/deliveryPipelines/web-app/releases/web-app-001/rollouts/web-app-001-to-test-0001
phases:
- deploymentJobs:
  deploymentJob:
    deploymentJob: {}
    id: deploy
    jobRun: projects/143823944593/locations/us-east1/deliveryPipelines/web-app/releases/web-app-001/rollouts/web-app-001-to-test-0001/jobRuns/b967eb05-f3e1-4806-85c1-344f37b559f7
    state: SUCCEEDED
    verifyJob:
      id: verify
      state: DISABLED
      verifyJob: {}
    id: stable
    state: SUCCEEDED
state: SUCCEEDED
targetId: test
uid: f657cf4351e84b6b97396c6a6dcea6c7
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5) $ kubectlx prod
kubectl get all -n web-app
Switched to context "prod".
NAME                                DESIRED  CURRENT  READY  AGE
replicaset.apps/leeroy-web-656d5c79ff 1         0        0      1s
student_04_6ad334ed764f@cloudshell:~/cloud-deploy-tutorials/tutorials/base (qwiklabs-gcp-04-93e61e9975d5) $
```

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Implement CI/CD Pipelines on Google Cloud

Working with Artifact Registry

Google Kubernetes Engine Pipeline using Cloud Build

Continuous Delivery with Google Cloud Deploy

DevSecOps > Course > Implement CI/CD Pipelines on Google Cloud >

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Lab instructions and tasks100/100

GSP1079

Overview

Objectives

Setup and requirements

Task 1. Set the variables

Task 2. Create three GKE clusters

Task 3. Prepare the web application container image

Task 4. Build and deploy the container images to

Implement CI/CD

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Checkpoints

Create three GKE clustersCheck my progress10 / 10

Create the web-app repositoryCheck my progress10 / 10

Build and deploy the container imagesCheck my progress10 / 10

Task 2. Create three GKE clusters

Task 3. Prepare the web application container image

Task 4. Build and deploy the container images to

Implement CI/CD

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Create the delivery pipeline

Check my progress10 / 10

Configure the deployment targets

Check my progress20 / 20

Create a release

Check my progress20 / 20

Promote the application to staging

Check my progress10 / 10

Task 2. Create three GKE clusters

Task 3. Prepare the web application container image

Task 4. Build and deploy the container images to

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Create a release

Check my progress20 / 20

Promote the application to staging

Check my progress10 / 10

Promote the application to prod

Check my progress10 / 10

Task 2. Create three GKE clusters

Task 3. Prepare the web application container image

Task 4. Build and deploy the container images to

Next

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